习题 1.1

1. (1) 抛三枚硬币 设样本空间Ω，正面为1，背面为0

$\Omega = \{(0,0,0), (1,0,0), (1,1,0), (1,1,1), (0,0,1), (0,1,1), (1,0,1), (0,1,0)\}$

(2) 抛三颗骰子 设样本空间Ω

$\Omega = \{(x,y,z) \mid x,y,z = 1,2,3,4,5,6\}$

(3) 抛一枚硬币，直至出现正面

$\Omega = \{(1), (0,1), (0,0,1), \dots\}$

(4) 黑Black 白White 红Red

$\Omega = \{(x,y) \mid x,y = B, W, R\}$

(5)

$\Omega = \{BW, BR, WB, WR, RB, RW\}$

1.1.7 (1) $A-(B-C) = (A-B)-C$

(3) $(A\cup B)-B = A$ 不成立

(4) $(A-B)\cup B = A$

(2) 若 ~~$AB\neq\emptyset$~~ $AB=\emptyset$ 且 $C\subset A$，则 $BC=\emptyset$

成立

1.1.10. (1) $A = AB\cup A\bar{B} = (B\cup\bar{B})\cap A = S\cap A = A$

~~$AB\cup A\bar{B} = (A\cup A)\cap B$~~ $B\cup\bar{B} = S$ $B\cap\bar{B} = \emptyset$

(2) $A\cup B = A\cup\bar{A}B$

$= A\cup(B\cap(A\cup\bar{A})) = A\cup(B\cap A)\cup(B\cap\bar{A})$

$B = (A\cap B)\cup(\bar{A}\cap B)$

应用概率与数理统计 陶志叶 2026年3月12日

习题1.2

4. 两颗骰子

(1) 点数之和为7

1+6, 2+4, ~~2+5~~ 3+3

设点数之和为6是事件A

$P(A) = 5(\frac{1}{6})^2 = \frac{5}{36}$

(2) 点数之和不超过6

1+4, 1+3, 1+2, 1+1, 1+5

2+4, 2+3, 2+2, ~~2+1~~ 1+1

3+3, ~~3+2~~ 3+2, ~~3+1~~ 3+1

4+1, 4+2

设点数之和不超过6是事件B

$P(B) = \cancel{2(\frac{1}{6}\times\frac{5}{6}) + 2(\frac{1}{6}\times\frac{5}{2}) + 2(\frac{1}{6}\times\frac{1}{3})} \frac{7}{6}$

$= \frac{10}{36} + \frac{2}{12} + \frac{1}{3} = \frac{28}{36} \to \frac{7}{9}$

$= \frac{15}{36} = \frac{5}{12}$

(3) 至少有一个6点

逆：一个6点都没有

设事件C为一个6点都没有

$P(C) = (1-\frac{1}{6})^2 = \frac{25}{36}$

故 $P(\bar{C}) = 1 - \frac{25}{36} = \frac{11}{36}$

11. 口袋十球

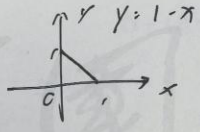
不放回任取四个

(1) 最小号码5

$\frac{C_1^1 C_5^3}{C_{10}^4} = \frac{1}{21}$

(2) $\frac{C_1^1 C_4^3}{C_{10}^4} = \frac{2}{105}$

27.



$2x = 1-x$

$x = \frac{1}{3}$

$S_1 = \frac{2}{3}\times 1\times\frac{1}{2} = \frac{1}{3}$

$S = 1\times 1\times\frac{1}{2} = \frac{1}{2}$

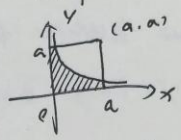
$P = \frac{S_1}{S} = \frac{2}{3}$

28. 设 $a>0$, $\forall x,y$

且 $0<x<a$, $0<y<a$

$xy < a^2/4$

~~正方形中不在反比例函数的部分~~



不对

样本空间 (x,y)

$D = \{(x,y) \mid 0<x<a, 0<y<a\}$

$4xy < a^2$

$x \in [\frac{a}{4}, a]$

$y \in [\frac{a^2}{4x}, a]$

$\int_{x=\frac{a}{4}}^{a} (a - \frac{a^2}{4x})\,dx$

$= a(a-\frac{a}{4}) - \frac{a^2}{4}\int_{\frac{a}{4}}^{a}\frac{1}{x}\,dx$

$= \frac{3a^2}{4} - \frac{a^2}{4}(\ln a - \ln\frac{a}{4})$

$P = 1 - (\frac{3a^2}{4} - \frac{a^2}{4}(\ln a - \ln\frac{a}{4}))$

$P = \frac{1}{4}(1+\ln 4)$

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